

```
In [17]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')
```

```
In [18]: file_path = 'sales_data_sample.csv'
df_csv = pd.read_csv(file_path, encoding='ISO-8859-1')

print("=== Loaded CSV Data ===")
print(df_csv.columns)
```

```
=== Loaded CSV Data ===
Index(['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH', 'ORDERLINENUMBER',
      'SALES', 'ORDERDATE', 'STATUS', 'QTR_ID', 'MONTH_ID', 'YEAR_ID',
      'PRODUCTLINE', 'MSRP', 'PRODUCTCODE', 'CUSTOMERNAME', 'PHONE',
      'ADDRESSLINE1', 'ADDRESSLINE2', 'CITY', 'STATE', 'POSTALCODE',
      'COUNTRY', 'TERRITORY', 'CONTACTLASTNAME', 'CONTACTFIRSTNAME',
      'DEALSIZE'],
      dtype='object')
```

```
In [19]: conn = sqlite3.connect("sales_data.db")
df_csv.to_sql("sales", conn, if_exists="replace", index=False)
```

Out[19]: 2823

```
In [20]: query = """
SELECT PRODUCTLINE AS product,
       SUM(QUANTITYORDERED) AS total_qty,
       ROUND(SUM(SALES), 2) AS revenue
FROM sales
GROUP BY PRODUCTLINE
"""

df_summary = pd.read_sql_query(query, conn)
conn.close()
```

```
In [21]: print("\n=== Sales Summary ===")
print(df_summary)
```

```
=== Sales Summary ===
```

	product	total_qty	revenue
0	Classic Cars	33992	3919615.66
1	Motorcycles	11663	1166388.34
2	Planes	10727	975003.57
3	Ships	8127	714437.13
4	Trains	2712	226243.47
5	Trucks and Buses	10777	1127789.84
6	Vintage Cars	21069	1903150.84

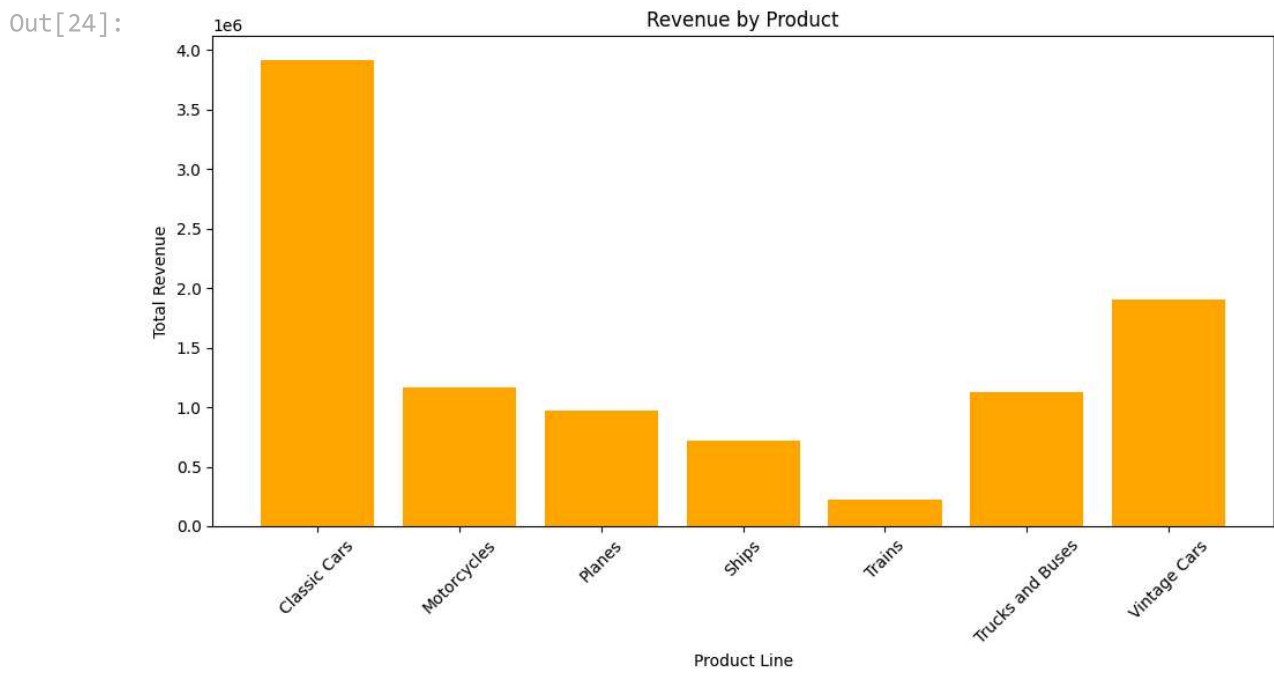
```
In [23]: plt.figure(figsize=(10, 6))
plt.bar(df_summary['product'], df_summary['revenue'], color='orange')
plt.title("Revenue by Product")
plt.xlabel("Product Line")
plt.ylabel("Total Revenue")
plt.xticks(rotation=45)
plt.tight_layout()

# Save instead of show
```

```
plt.savefig("revenue_by_product.png")  
print("Plot saved as 'revenue_by_product.png'")
```

Plot saved as 'revenue_by_product.png'

```
In [24]: from IPython.display import Image  
Image("revenue_by_product.png")
```



In []:

In []: